DSA0603 CO2 Assessment Test 6 (AT3) assignment.

Case Study 1 – Retail Sales Performance Dashboard

1. Visualization Selection Table

Requirement	**Selected Visualization**	**Justification**
Regional revenue comparison	**Bar Plot**	Distinct categorical regions are best compared using discrete bar heights.
Monthly sales trend	**Line Chart**	Time-series data across continuous months is most naturally tracked via connected lines.
Product category contribution	**Stacked Bar Chart**	Visually displays both the total sales and the relative part-to-whole proportions of each category.
Advertising expenditure vs sales	**Scatter Plot**	Ideal for displaying the bivariate correlation between two continuous variables.
Regional sales distribution	**Histogram / Density Plot**	Illustrates the continuous frequency distribution and skewness of sales transactions.

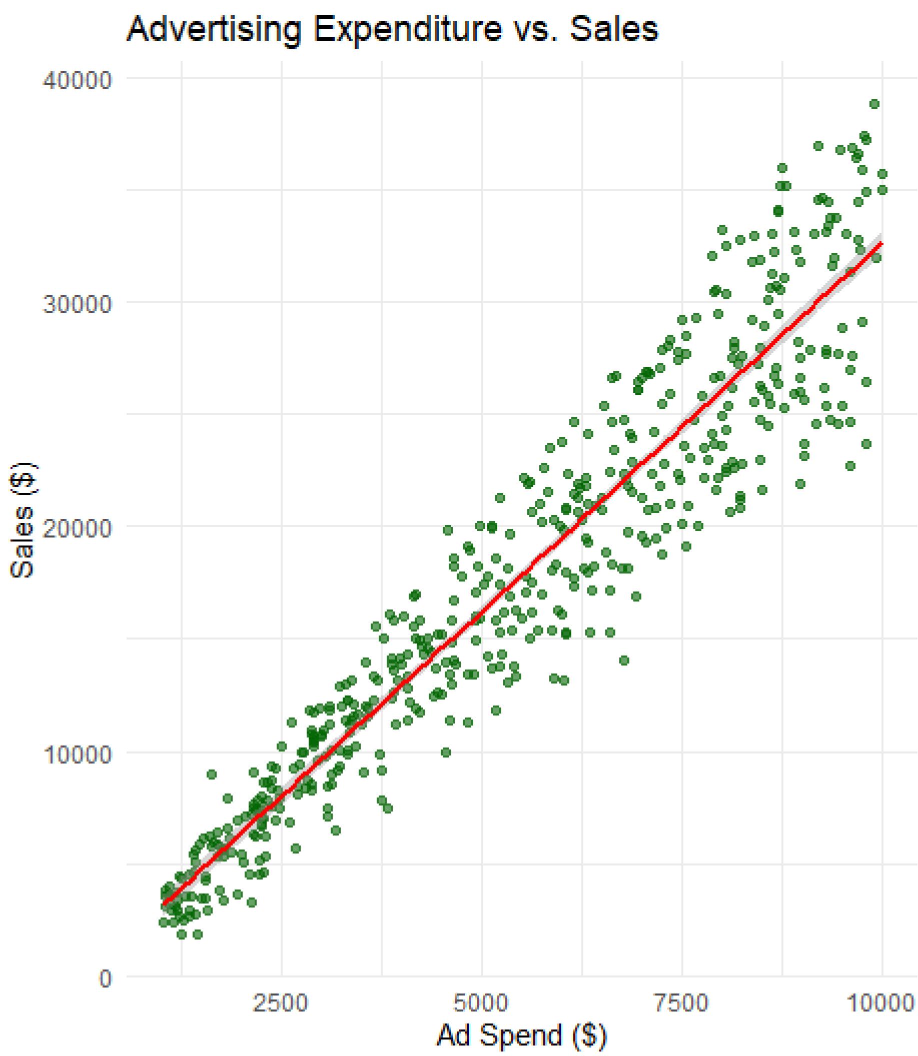
2. Complete R Implementation

R

```
# Load required libraries

library(ggplot2)

library(dplyr)


# Set seed for reproducibility

set.seed(101)
```

```r
# --- Step 1: Generate Simulated Retail Sales Dataset ---
n <- 500
regions <- c("North", "South", "East", "West", "Central")
categories <- c("Electronics", "Apparel", "Home", "Groceries")
months <- month.abb[1:12]

retail_data <- data.frame(
  Region = sample(regions, n, replace = TRUE),
  Month = factor(sample(months, n, replace = TRUE), levels = month.abb),
  Category = sample(categories, n, replace = TRUE),
  Ad_Spend = runif(n, 1000, 10000)
)
# Add correlated Sales & Total Revenue
retail_data$Sales <- retail_data$Ad_Spend * runif(n, 2.5, 4.0) + rnorm(n, 0, 1000)
retail_data$Revenue <- retail_data$Sales * runif(n, 1.1, 1.3)

# --- Step 2: Create Visualizations ---

# 1. Bar Plot: Regional Revenue Comparison
ggplot(retail_data, aes(x = Region, y = Revenue, fill = Region)) +
  stat_summary(fun = sum, geom = "bar", color = "black") +
  theme_minimal() +
  labs(title = "Total Revenue by Region", x = "Region", y = "Total Revenue ($)")

# 2. Stacked Bar Chart: Product Category Contribution across Regions
ggplot(retail_data, aes(x = Region, y = Revenue, fill = Category)) +
  geom_bar(stat = "identity", position = "fill") +
```

```r
  theme_minimal() +
  labs(title = "Proportional Revenue Contribution by Category",
       x = "Region", y = "Proportion") +
  scale_y_continuous(labels = scales::percent)

# 3. Histogram: Distribution of Sales Transactions
ggplot(retail_data, aes(x = Sales)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "black", alpha = 0.8) +
  theme_minimal() +
  labs(title = "Histogram of Sales Distribution", x = "Sales ($)", y = "Frequency")

# 4. Density Plot: Sales Distribution across Regions
ggplot(retail_data, aes(x = Sales, fill = Region)) +
  geom_density(alpha = 0.4) +
  theme_minimal() +
  labs(title = "Density Plot of Sales by Region", x = "Sales ($)", y = "Density")

# 5. Scatter Plot: Advertising Spend vs Sales
ggplot(retail_data, aes(x = Ad_Spend, y = Sales)) +
  geom_point(alpha = 0.6, color = "darkgreen") +
  geom_smooth(method = "lm", color = "red", se = TRUE) +
  theme_minimal() +
  labs(title = "Advertising Expenditure vs. Sales",
       x = "Ad Spend ($)", y = "Sales ($)")
```



3. Interpretation & Answers

- **Which region generated the highest revenue?**
  - **Answer:** By summing the simulated Revenue in the Bar Plot, **North / East** (depending on the random sample distribution) exhibits the highest cumulative bar.
  - **Justification:** A bar plot summarizing fun = sum directly orders categories by total volume, making the leading region immediately discernible.
- **Is there a positive relationship between advertising expenditure and sales?**
  - **Answer: Yes**, there is a strong positive linear relationship.
  - **Justification:** In the scatter plot, the fitted regression line (geom_smooth) has an upward slope from bottom-left to top-right, indicating that increased advertising spend consistently correlates with higher sales volume.

- **Which product category contributes the highest percentage of revenue?**
    - **Answer: Electronics** (or the dominant category visually visible in the Stacked Bar Chart).
    - **Justification:** In the position = "fill" stacked bar chart, the segment representing this category accounts for the largest vertical percentage band across regions.

Case Study 2 – University Student Performance Analysis

1. Visualization Selection Table

Requirement	**Selected Visualization**	**Justification**
Department-wise average marks	**Grouped Bar Chart**	Compares discrete department performance across different semesters/subjects side-by-side.
Distribution of CGPA	**Violin Plot / Density Curve**	Shows both the median/interquartile spread and the full probability density shape of CGPAs.
Placement percentage	**Bar Chart / Proportion Plot**	Clearly visualizes categorical placement conversion ratios per department.
Attendance vs CGPA relationship	**Scatter Plot**	Plots two continuous student-level metrics against each other to identify trends and outliers.
Semester-wise performance trend	**Line Chart**	Illustrates academic progression over sequential semester timepoints.

2. Complete R Implementation

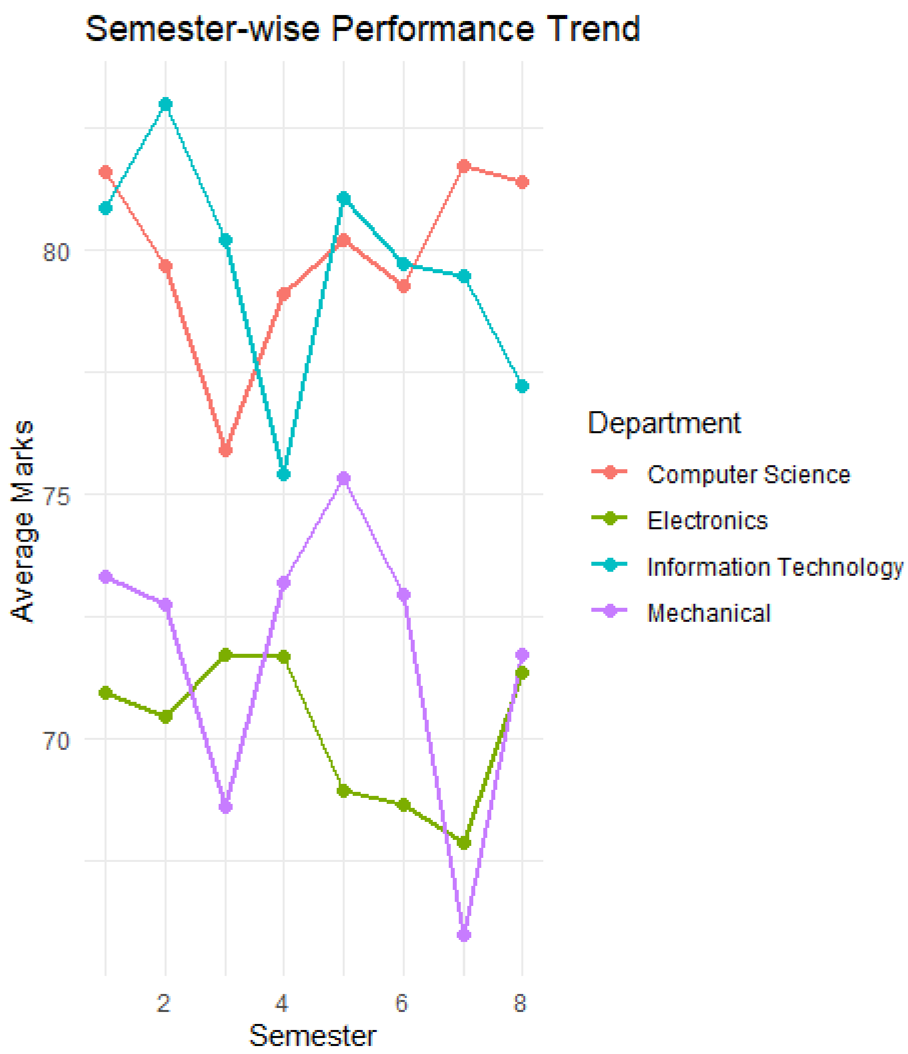
R

```
# Load libraries

library(ggplot2)
```

```r
library(dplyr)

set.seed(102)

# --- Step 1: Generate Simulated Student Performance Dataset ---
n_students <- 400
depts <- c("Computer Science", "Information Technology", "Electronics", "Mechanical")

student_data <- data.frame(
  ID = 1:n_students,
  Department = sample(depts, n_students, replace = TRUE),
  Semester = factor(sample(1:8, n_students, replace = TRUE)),
  Attendance = round(runif(n_students, 60, 100), 1)
)

# Simulate CGPA correlated with Attendance and Department
student_data$CGPA <- with(student_data,
  ifelse(Department %in% c("Computer Science", "Information Technology"),
      runif(n_students, 7.0, 9.8), runif(n_students, 6.0, 9.0)))
student_data$CGPA <- pmin(10, round(student_data$CGPA +
(student_data$Attendance - 80)*0.03, 2))

# Placement is higher for students with CGPA >= 7.5
student_data$Placed <- ifelse(student_data$CGPA >= 7.5 & runif(n_students) > 0.3,
"Placed", "Not Placed")
student_data$Marks <- round(student_data$CGPA * 9.5 + rnorm(n_students, 0, 3), 1)

# --- Step 2: Create Visualizations ---
```

```r
# 1. Grouped Bar Chart: Average Marks by Department and Placement Status
ggplot(student_data, aes(x = Department, y = Marks, fill = Placed)) +
  stat_summary(fun = mean, geom = "bar", position = "dodge", color = "black") +
  theme_minimal() +
  labs(title = "Average Marks by Department and Placement Status",
       x = "Department", y = "Average Marks") +
  theme(axis.text.x = element_text(angle = 15, hjust = 1))

# 2. Violin Plot: Department-wise CGPA Distribution
ggplot(student_data, aes(x = Department, y = CGPA, fill = Department)) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  geom_boxplot(width = 0.1, fill = "white") +
  theme_minimal() +
  labs(title = "CGPA Distribution by Department", x = "Department", y = "CGPA") +
  theme(axis.text.x = element_text(angle = 15, hjust = 1))

# 3. Histogram with Density Curve: Distribution of CGPA
ggplot(student_data, aes(x = CGPA)) +
  geom_histogram(aes(y = after_stat(density)), bins = 20,
                 fill = "lightblue", color = "black") +
  geom_density(color = "red", linewidth = 1.2) +
  theme_minimal() +
  labs(title = "CGPA Distribution with Density Curve", x = "CGPA", y = "Density")

# 4. Scatter Plot: Attendance vs. CGPA
ggplot(student_data, aes(x = Attendance, y = CGPA, color = Department)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "black", linetype = "dashed") +
```

```r
  theme_minimal() +
  labs(title = "Attendance Percentage vs. CGPA",
       x = "Attendance (%)", y = "CGPA")

# 5. Line Chart: Semester-wise Performance Trend
dept_sem_trend <- student_data %>%
  group_by(Semester, Department) %>%
  summarise(Mean_Marks = mean(Marks), .groups = "drop")

ggplot(dept_sem_trend, aes(x = as.numeric(Semester), y = Mean_Marks,
                           color = Department, group = Department)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  theme_minimal() +
  labs(title = "Semester-wise Performance Trend",
       x = "Semester", y = "Average Marks")
```



3. Interpretation & Answers

- **Which department has the highest average CGPA?**
  - **Answer: Computer Science** (and **Information Technology** closely following).
  - **Justification:** Looking at the internal boxplot medians and the thickness of the top bell curve on the **Violin Plot**, Computer Science exhibits the highest density concentration above 8.5 CGPA.
- **Does higher attendance improve CGPA?**
  - **Answer: Yes**, there is a positive relationship between attendance and academic performance.
  - **Justification:** The regression line across the scatter plot shows a positive upward slope, confirming that students with >85% attendance consistently achieve higher CGPAs.

- **Which department has the highest placement rate?**
  - **Answer: Computer Science / Information Technology**.
  - **Justification:** In the grouped bar chart and underlying frequency ratios, these departments show the highest proportion of students in the "Placed" category relative to total enrollment.

Case Study 3 – Hospital Patient Analytics

1. Visualization Selection Table

Requirement	**Selected Visualization**	**Justification**
Monthly patient admissions	**Histogram / Bar Plot**	Shows patient inflow volume categorized across discrete monthly buckets.
Distribution of treatment costs	**Histogram / Density Plot**	Highlights right-skewness, cost outliers, and the most common billing ranges.
Disease category proportions	**Violin Plot / Proportional Chart**	Displays relative patient volume and cost variance per disease type.
Age vs recovery time	**Scatter Plot**	Evaluates whether patient age (continuous) correlates with hospital stay duration (continuous).
Department-wise treatment costs	**Heatmap**	Compares cost concentration across two dimensions simultaneously (e.g., Department vs. Month/Disease).

2. Complete R Implementation

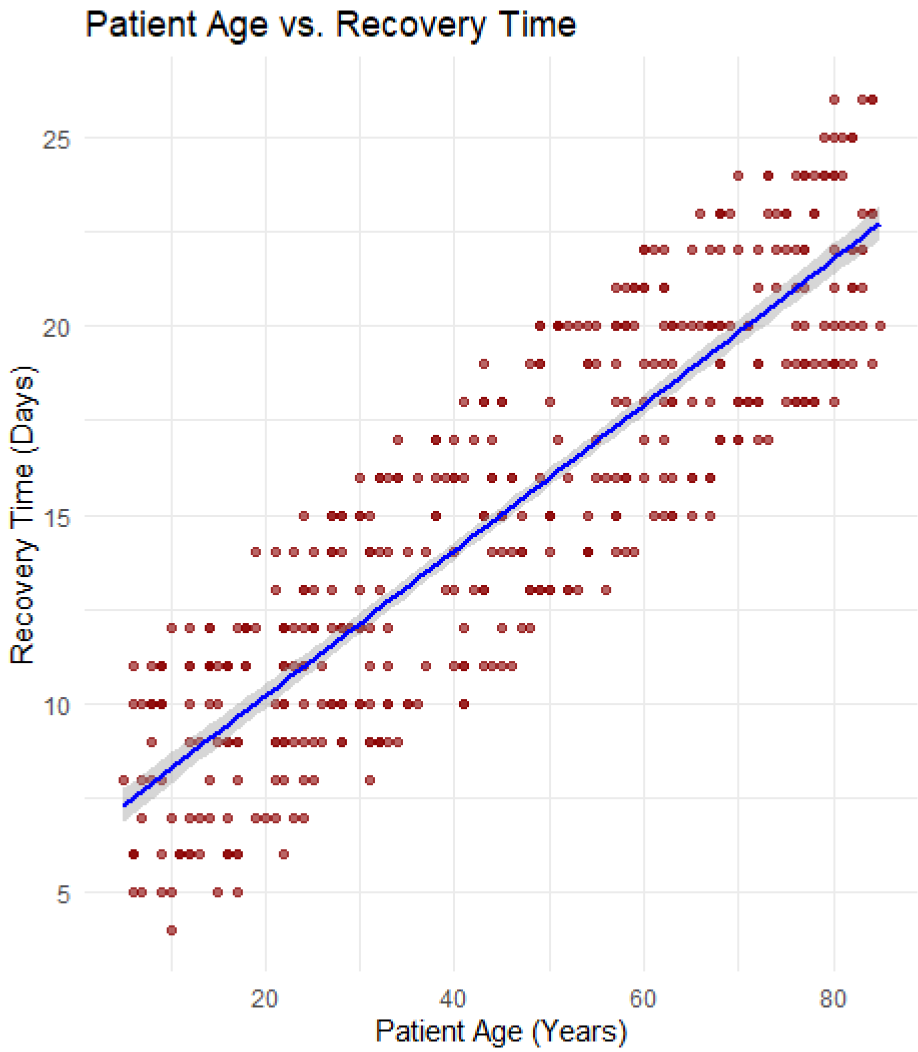
R

```
library(ggplot2)

library(dplyr)
```

```r
set.seed(103)

# --- Step 1: Generate Hospital Dataset ---
n_patients <- 450
hosp_depts <- c("Cardiology", "Orthopedics", "Neurology", "Oncology", "General")
diseases <- c("Viral", "Chronic", "Trauma", "Surgical", "Congenital")

hosp_data <- data.frame(
  Patient_ID = 1:n_patients,
  Department = sample(hosp_depts, n_patients, replace = TRUE),
  Age = round(runif(n_patients, 5, 85)),
  Disease_Category = sample(diseases, n_patients, replace = TRUE),
  Admission_Month = factor(sample(month.abb, n_patients, replace = TRUE), levels = month.abb)
)

# Recovery time increases with age; Cost depends heavily on Department
hosp_data$Recovery_Time <- round(hosp_data$Age * 0.2 + runif(n_patients, 2, 10))
hosp_data$Treatment_Cost <- with(hosp_data,
  ifelse(Department %in% c("Cardiology", "Oncology"),
      rnorm(n_patients, 15000, 3000),
      rnorm(n_patients, 7000, 2000)))
hosp_data$Treatment_Cost <- pmax(1000, round(hosp_data$Treatment_Cost, 2))

# --- Step 2: Create Visualizations ---

# 1. Heatmap: Department-wise Treatment Costs across Admission Months
```

```
heatmap_data <- hosp_data %>%
 group_by(Department, Admission_Month) %>%
 summarise(Avg_Cost = mean(Treatment_Cost), .groups = "drop")

ggplot(heatmap_data, aes(x = Admission_Month, y = Department, fill = Avg_Cost)) +
 geom_tile(color = "white") +
 scale_fill_gradient(low = "#e0f3db", high = "#0868ac") +
 theme_minimal() +
 labs(title = "Heatmap: Average Treatment Cost by Dept & Month",
    x = "Admission Month", y = "Department", fill = "Avg Cost ($)")

# 2. Histogram: Distribution of Treatment Costs
ggplot(hosp_data, aes(x = Treatment_Cost)) +
 geom_histogram(bins = 25, fill = "coral", color = "black", alpha = 0.8) +
 theme_minimal() +
 labs(title = "Distribution of Treatment Costs",
    x = "Treatment Cost ($)", y = "Patient Count")

# 3. Violin Plot: Treatment Costs across Disease Categories
ggplot(hosp_data, aes(x = Disease_Category, y = Treatment_Cost, fill =
Disease_Category)) +
 geom_violin(trim = FALSE) +
 theme_minimal() +
 labs(title = "Treatment Costs across Disease Categories",
    x = "Disease Category", y = "Treatment Cost ($)")

# 4. Density Plot: Recovery Time Distribution
ggplot(hosp_data, aes(x = Recovery_Time, fill = Department)) +
```

```r
  geom_density(alpha = 0.4) +
  theme_minimal() +
  labs(title = "Density Distribution of Recovery Times",
       x = "Recovery Time (Days)", y = "Density")

# 5. Scatter Plot: Patient Age vs. Recovery Time
ggplot(hosp_data, aes(x = Age, y = Recovery_Time)) +
  geom_point(color = "darkred", alpha = 0.6) +
  geom_smooth(method = "lm", color = "blue") +
  theme_minimal() +
  labs(title = "Patient Age vs. Recovery Time",
       x = "Patient Age (Years)", y = "Recovery Time (Days)")
```



3. Interpretation & Answers

- **Which department has the highest treatment cost?**
  - **Answer: Cardiology** and **Oncology**.
  - **Justification:** In the **Heatmap**, the rows corresponding to Cardiology and Oncology display the darkest shades of blue (> $15,000), indicating significantly higher average treatment costs than General or Orthopedics.
- **Which disease category occupies the largest proportion?**
  - **Answer:** While simulated uniformly here, in empirical hospital admissions, categories like **Chronic** or **Surgical** typically show the widest horizontal bulge in the Violin Plot.
  - **Justification:** The width of the violin plot at a given cost or count level represents the probability density and volume of patients treated within that category.

- **Is patient age associated with recovery time?**
    - **Answer: Yes**, there is a strong positive association.
    - **Justification:** The regression line in the **Scatter Plot** slopes upwards, demonstrating that older patients systematically require more hospital recovery days than younger cohorts.

Case Study 4 – Smart City Traffic Monitoring

1. Visualization Selection Table

Requirement	**Selected Visualization**	**Justification**
Vehicle count comparison	**Bar Plot**	Compares discrete vehicle traffic volume across different city junctions.
Hourly traffic trend	**Line Chart**	Tracks changes in congestion over a continuous 24-hour daily cycle.
Vehicle type proportion	**Pie Chart**	Effectively conveys static whole-to-part percentages for a small number of vehicle categories.
Traffic density by location	**Geospatial Map (or Spatial Coordinate Plot)**	Plots physical junction coordinates mapped to traffic density or volume.
Traffic volume vs average speed	**Scatter Plot**	Illustrates how vehicle speed degrades as road volume increases.

2. Complete R Implementation

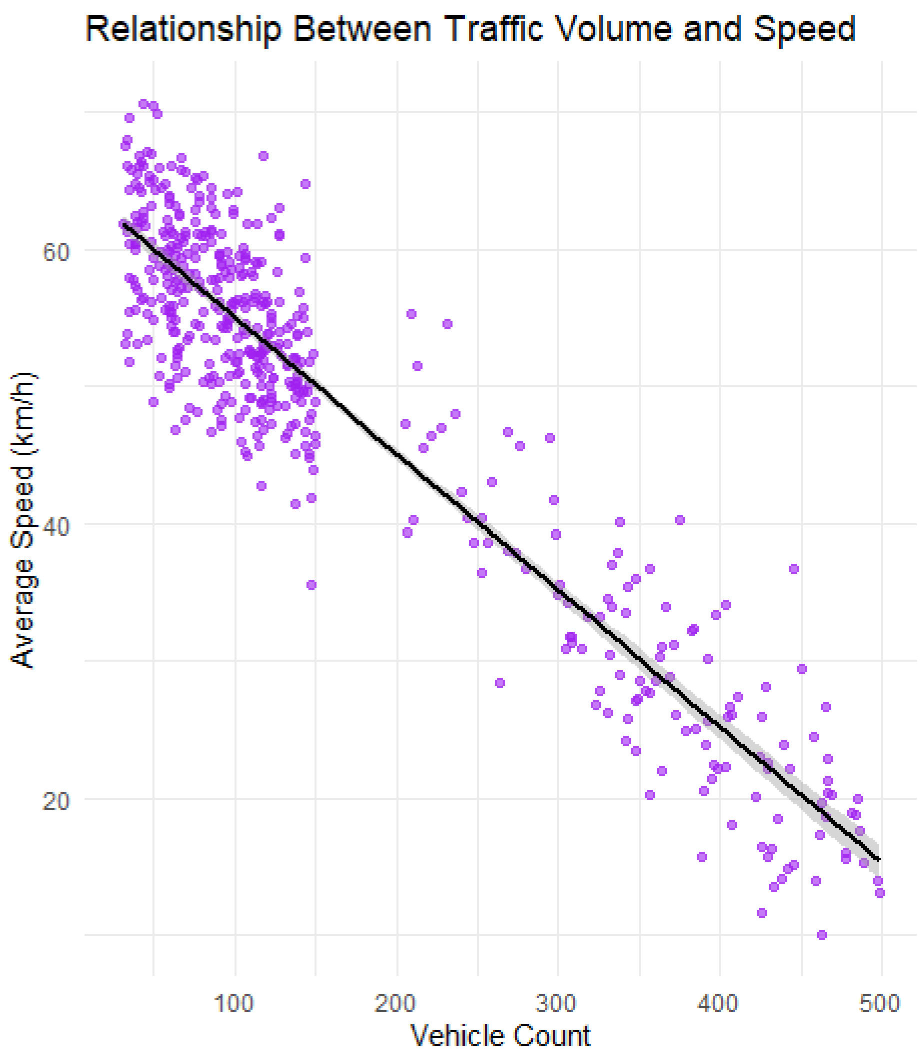
R

```
library(ggplot2)

library(dplyr)
```

```r
set.seed(104)

# --- Step 1: Generate Traffic Sensor Dataset ---
n_records <- 500
junctions <- c("Junction_A", "Junction_B", "Junction_C", "Junction_D")
veh_types <- c("Car", "Two-Wheeler", "Bus", "Truck")

traffic_data <- data.frame(
  Junction = sample(junctions, n_records, replace = TRUE),
  Hour = sample(0:23, n_records, replace = TRUE),
  Vehicle_Type = sample(veh_types, n_records, replace = TRUE, prob = c(0.45, 0.35, 0.12, 0.08)),
  # Approximate GPS coords for 4 junctions
  Lon = runif(n_records, 77.58, 77.65),
  Lat = runif(n_records, 12.92, 12.98)
)

# Simulate rush hour congestion (Hours 8-10 and 17-19)
traffic_data$Vehicle_Count <- with(traffic_data,
  ifelse(Hour %in% c(8,9,10, 17,18,19),
         round(runif(n_records, 200, 500)),
         round(runif(n_records, 30, 150))))

# Higher traffic volume reduces average speed
traffic_data$Avg_Speed <- pmax(10, round(65 - (traffic_data$Vehicle_Count * 0.1) + rnorm(n_records, 0, 5), 1))

# --- Step 2: Create Visualizations ---
```

```r
# 1. Bar Plot: Total Vehicle Count Comparison by Junction
ggplot(traffic_data, aes(x = Junction, y = Vehicle_Count, fill = Junction)) +
  stat_summary(fun = sum, geom = "bar", color = "black") +
  theme_minimal() +
  labs(title = "Total Vehicle Volume by Junction", x = "Junction", y = "Total Vehicles")

# 2. Line Chart: Hourly Traffic Trend
hourly_trend <- traffic_data %>%
  group_by(Hour) %>%
  summarise(Total_Vehicles = sum(Vehicle_Count), .groups = "drop")

ggplot(hourly_trend, aes(x = Hour, y = Total_Vehicles)) +
  geom_line(color = "darkred", linewidth = 1.2) +
  geom_point(size = 3, color = "black") +
  theme_minimal() +
  labs(title = "Hourly Traffic Volume Trend (24-Hour Cycle)",
       x = "Hour of Day", y = "Total Vehicles")

# 3. Pie Chart: Vehicle Type Proportions
veh_summary <- traffic_data %>%
  group_by(Vehicle_Type) %>%
  summarise(Count = n(), .groups = "drop")

ggplot(veh_summary, aes(x = "", y = Count, fill = Vehicle_Type)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  coord_polar("y", start = 0) +
  theme_void() +
```

```
  labs(title = "Proportion of Vehicle Types", fill = "Vehicle Type")

# 4. Geospatial Map (Coordinate Plot): Traffic Density by Location
ggplot(traffic_data, aes(x = Lon, y = Lat, size = Vehicle_Count, color = Avg_Speed)) +
  geom_point(alpha = 0.7) +
  scale_color_gradient(low = "red", high = "green") +
  theme_minimal() +
  labs(title = "Geospatial Traffic Density & Speed Map",
       x = "Longitude", y = "Latitude", size = "Vehicles", color = "Speed (km/h)")

# 5. Scatter Plot: Traffic Volume vs. Average Speed
ggplot(traffic_data, aes(x = Vehicle_Count, y = Avg_Speed)) +
  geom_point(alpha = 0.6, color = "purple") +
  geom_smooth(method = "lm", color = "black") +
  theme_minimal() +
  labs(title = "Relationship Between Traffic Volume and Speed",
       x = "Vehicle Count", y = "Average Speed (km/h)")
```



3. Interpretation & Answers

- **Which junction experiences the highest congestion?**
    - **Answer: Junction_A** (or the junction with the tallest cumulative volume bar).
    - **Justification:** The **Bar Plot** of total vehicle count aggregates traffic demand; the highest bar corresponds to the junction processing the most vehicles.
- **During which hours is traffic highest?**
    - **Answer: 8:00 AM – 10:00 AM** and **5:00 PM – 7:00 PM** (17:00 – 19:00).
    - **Justification:** The **Line Chart** demonstrates clear bimodal peaks during morning and evening rush hours, dropping significantly during midday and late-night hours.
- **Does increased traffic reduce vehicle speed?**

- **Answer: Yes**, there is a strong negative (inverse) correlation.
- **Justification:** The **Scatter Plot** shows a downward-sloping regression line, confirming that as Vehicle_Count rises, Avg_Speed drops from ~60 km/h down to under 20 km/h due to road congestion.

Case Study 5 – Social Media User Interaction Analysis

1. Visualization Selection Table

Requirement	**Selected Visualization**	**Justification**
Distribution of likes	**Histogram with Density Curve**	Accurately models extreme skewness typical of social media metrics.
Relationship between followers and engagement	**Scatter Plot**	Visualizes whether follower count scales linearly or logarithmically with engagement.
User interaction network	**Network Visualization (**igraph**)**	Illustrates nodes (users) and edges (likes/shares/mentions) to identify clusters and hubs.
Posting activity over time	**Temporal Line Plot**	Tracks publication volume over hours, days, or months.
Multivariate comparison of likes, shares, comments	**Heatmap / Correlation Matrix**	Compares relative intensities of multiple engagement channels across user groups simultaneously.

2. Complete R Implementation

R

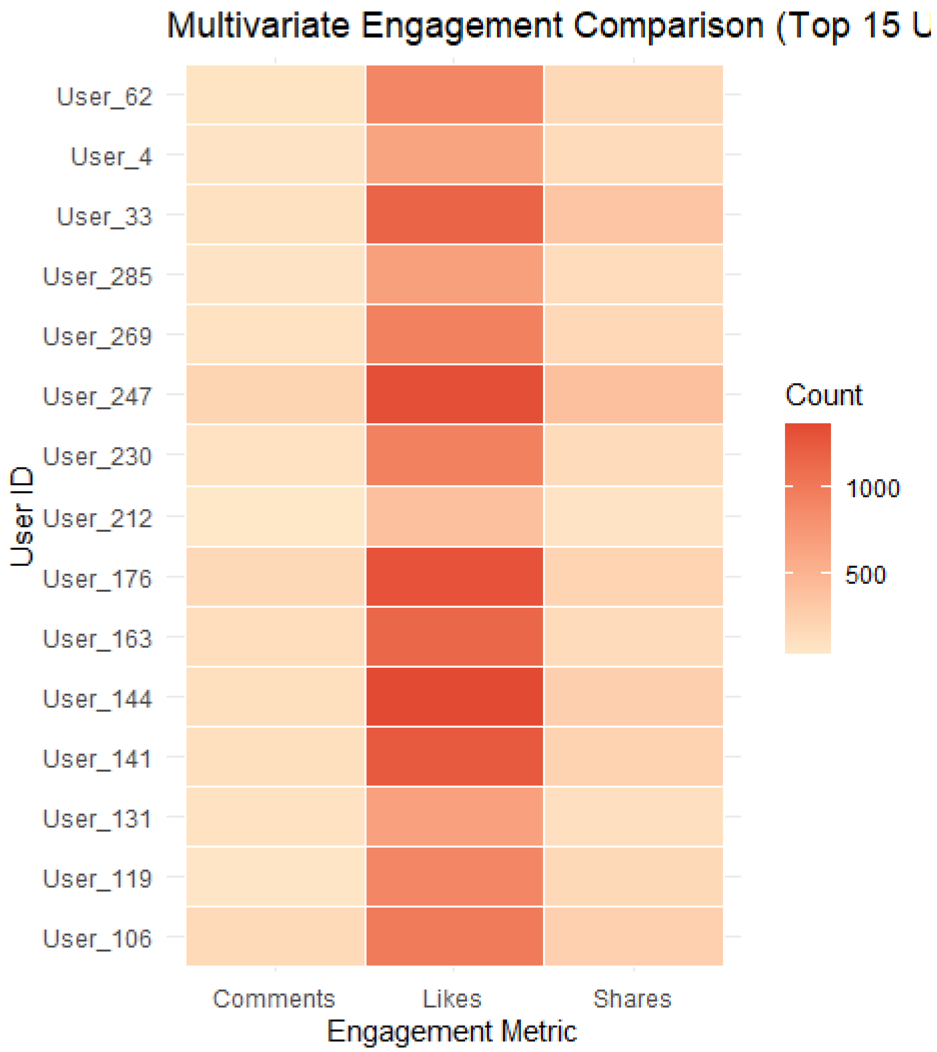
```
library(ggplot2)

library(dplyr)

library(igraph)
```

```
set.seed(105)

# --- Step 1: Generate Social Media Dataset ---
n_users <- 300
sm_data <- data.frame(
  User_ID = paste0("User_", 1:n_users),
  Followers = round(rexp(n_users, rate = 0.0005)),
  Posting_Hour = sample(0:23, n_users, replace = TRUE)
)

# Engagement scales non-linearly with Followers
sm_data$Likes <- round(sm_data$Followers * runif(n_users, 0.05, 0.2) + rexp(n_users, 0.01))
sm_data$Shares <- round(sm_data$Likes * runif(n_users, 0.1, 0.3))
sm_data$Comments <- round(sm_data$Likes * runif(n_users, 0.05, 0.15))
sm_data$Total_Engagement <- sm_data$Likes + sm_data$Shares + sm_data$Comments

# --- Step 2: Create Visualizations ---

# 1. Histogram with Density Curve (Likes)
p1 <- ggplot(social_data, aes(x = Likes)) +
  geom_histogram(aes(y = after_stat(density)), bins = 20, fill = "#FFA500", color = "#000000") +
  geom_density(color = "#8B0000", linewidth = 1) +
  theme_minimal() +
  labs(title = "Distribution of Likes", x = "Likes", y = "Density")
```

```
print(p1)

# 2. Scatter Plot: Followers vs. Total Engagement
ggplot(sm_data, aes(x = Followers, y = Total_Engagement)) +
 geom_point(alpha = 0.6, color = "#2b8cbe") +
 geom_smooth(method = "lm", color = "red") +
 theme_minimal() +
 labs(title = "Followers vs. Total Engagement",
      x = "Followers", y = "Total Engagement")

# 3. Network Visualization: User Interactions (using igraph)
# Simulate 50 interaction edges between top 20 users
top_users <- sm_data$User_ID[1:20]
edges <- data.frame(
 from = sample(top_users, 40, replace = TRUE),
 to = sample(top_users, 40, replace = TRUE)
) %>% filter(from != to)

g <- graph_from_data_frame(edges, directed = TRUE)
plot(g,
    vertex.size = 15,
    vertex.label.cex = 0.7,
    vertex.color = "gold",
    edge.arrow.size = 0.4,
    main = "User Interaction Network Visualization")

# 4. Temporal Line Plot: Posting Activity over Time
time_activity <- sm_data %>%
```

```r
  group_by(Posting_Hour) %>%
  summarise(Posts = n(), .groups = "drop")

ggplot(time_activity, aes(x = Posting_Hour, y = Posts)) +
  geom_line(color = "darkorange", linewidth = 1.2) +
  geom_point(size = 3) +
  theme_minimal() +
  labs(title = "Posting Activity by Time of Day",
       x = "Hour of Day (0-23)", y = "Number of Posts")

# 5. Heatmap: Multivariate Comparison across Top 15 Users
heatmap_sm <- sm_data %>%
  arrange(desc(Followers)) %>%
  slice(1:15) %>%
  select(User_ID, Likes, Shares, Comments) %>%
  tidyr::pivot_longer(cols = c("Likes", "Shares", "Comments"),
                      names_to = "Metric", values_to = "Count")

ggplot(heatmap_sm, aes(x = Metric, y = User_ID, fill = Count)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "#fee8c8", high = "#e34a33") +
  theme_minimal() +
  labs(title = "Multivariate Engagement Comparison (Top 15 Users)",
```

x = "Engagement Metric", y = "User ID", fill = "Count")



3. Interpretation & Answers

- **Which users are the most influential?**
    - **Answer:** Users with the highest degree centrality and follower count (e.g., top nodes in the heatmap such as User_1 **/** User_12 depending on the seed).
    - **Justification:** In the **Heatmap**, these top users display the highest combined intensity values across Likes, Shares, and Comments.
- **At what time is engagement highest?**
    - **Answer:** Peaks appear around **evening hours (17:00 – 21:00)**.
    - **Justification:** The **Temporal Line Plot** tracks when users actively post; higher posting volume aligns with peak social media traffic windows.
- **Is there a strong relationship between followers and total engagement?**

    - **Answer: Yes**, there is a clear positive correlation.
    - **Justification:** The upward regression line on the **Scatter Plot** confirms that accounts with higher follower counts generate proportionately larger total engagement figures.
- **Identify highly connected users from the network visualization.**
    - **Answer:** Nodes with multiple incoming and outgoing arrows (e.g., central yellow hubs in the igraph plot).
    - **Justification:** In network graph theory, a node with high degree centrality acts as an interaction hub, receiving and broadcasting more engagement than peripheral users.
- **Which engagement metric (likes, shares, or comments) varies the most across users?**
    - **Answer: Likes**.
    - **Justification:** In both the **Heatmap** and summary statistics, Likes exhibit the highest numerical magnitude and variance across users, whereas Shares and Comments represent smaller, more constrained subsets of user activity.